

Make your cabinet

You can use a variety of materials such as plywood and sheet metal to house your amplifier. You can also use ready-made cabinets available in electronic stores

Shielding

When using cables for audio signals, always use coaxial cables with a braided outer layer. This braided layer is grounded and prevents external noise interference

Resistor colour code

The sentence – **BB Roy** (from) **Great Britain** (had a) **Very Good Wife** – is commonly used as a mnemonic for the colour code. The first letter of each word corresponds to the colour band.

Colour code	Value
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Gray	8
White	9

Accordingly, if the resistor is colour coded Red-Red-Orange, then its value is $22 \times 10^3 = 22 \times 1000 = 22 \text{ k}$.

signals are out-of-phase. Electronic circuitry need a DC supply to work. Effectively, this is where our first circuit stage is needed. A rectifier circuit converts the lower voltage alternating current to a more stabilised direct current (DC) supply. There are two kinds of rectifiers – half wave and full wave rectifiers. The most common type of rectifier circuit is the bridge circuit which is nothing but four diodes (1N4007) connected as a bridge. Input AC voltage is applied across two ends of the bridge, while unregulated rectified DC voltage from the other ends are fed to the voltage regulator stage. The voltage regulator is built around positive and negative voltage regulators ICs 7815 and 7915, respectively. The one tricky thing about these voltage regulators is their pin outs. Depending on how you look at them, pin numbers 1 to 3 are arranged differently. Test your connections

separately before attempting to power your amplifier off it. With all going right, the output voltage should be stable at +15 V and -15 V, respectively.

With your power supply done, the most important part is the amplifier itself. For this, a closer look at the circuit diagram will tell you how to go about building the circuit. The two ICs as explained earlier are two stages used to process the same audio signal. In fact, both the stages are wired identically. The only difference is the output audio signal that is fed back to the inverting input of IC2 via the RC stage comprising R7 and C5. And yes, the +15V and -15V that you got from the regulator stage will be connected to the points marked +Vs and -Vs. Similarly, the point marked with the black line (GND) is connected to the centre tap of the transformer (also the common point of ICs 7815 and 7915).

Stages

The circuit is made up of several stages. A look at Figure 1 will give you a fair idea of how your amplifier will work.

Input stage

Audio signal is input via a 3.5-mm audio jack. Connect

Reading capacitor values

Ceramic capacitors have a numbering scheme instead of colours. The first number represents the first digit, the second number, the second digit, the third number is the number of zeros to give the capacitance in pF. For example, 102 indicates 10×10^2 in pF = $10 \times 100 = 1000 \text{ pF} = 1 \text{ nF}$

the wire soldered to the outer connector to the ground terminal, and the inner wire carries the audio signal. Capacitor C1 couples the audio to the input of the amplifier and reduces the DC level at the input of the amplifier.

Biasing/Feedback

Resistors R1 and R2 (22k) are connected to pin 1 of TDA2030, respectively. Also, R3, R4 and C4; and R5, C5 and R6 together form the biasing network for ICs 1 and 2. R7 couples the audio between IC1 and IC2.

Output stage

Capacitors C8, R8, C9 and R9 form the output stage before interfacing the speaker. A series RC-circuit (resistor and capacitor connected one after the other) helps boost the audio signal. Use a speaker connector to connect your speakers here.

Power supply stage

Pin 5 of IC TDA 2030A is connected to +15V and pin 3 to -15V. Capacitors C6, C7, C2 and C3 form the basic by-pass circuitry to remove AC components from the regulated power supply. This minimises AC hum due to 50 Hz mains AC supply voltage.

Soldering

File the component leads to get rid of oxide deposits. This will aid a good solder joint. Insert the component into the PCB and on the other end, place the hot soldering iron tip across the component lead near the copper base of the board, while applying the solder alloy. 2-3 seconds should be ideal.

Powering on

Before switching on your amplifier, check all the connections with the continuity checker on a multimeter, ensuring they're all connected. If all seems fine, your amplifier should work, unless a particular component is faulty; in which case, you'd have to isolate and replace the faulty component. If you've followed the circuit diagram, your amplifier should function properly; and yeah, as far as the price goes, you should be done within Rs. 250. **d**

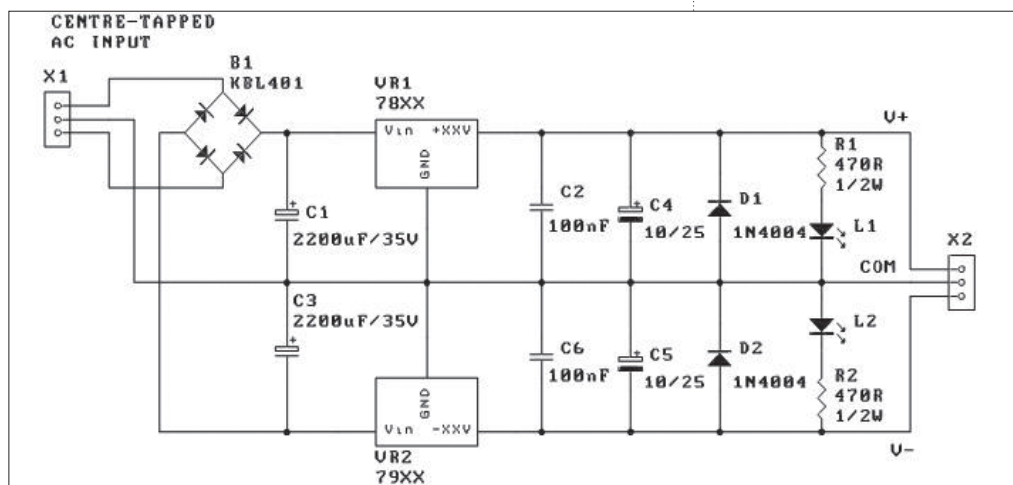


Figure 2: You need to convert AC to DC to power the amplifier circuit in Figure 1. Use 78XX series voltage regulator ICs for this